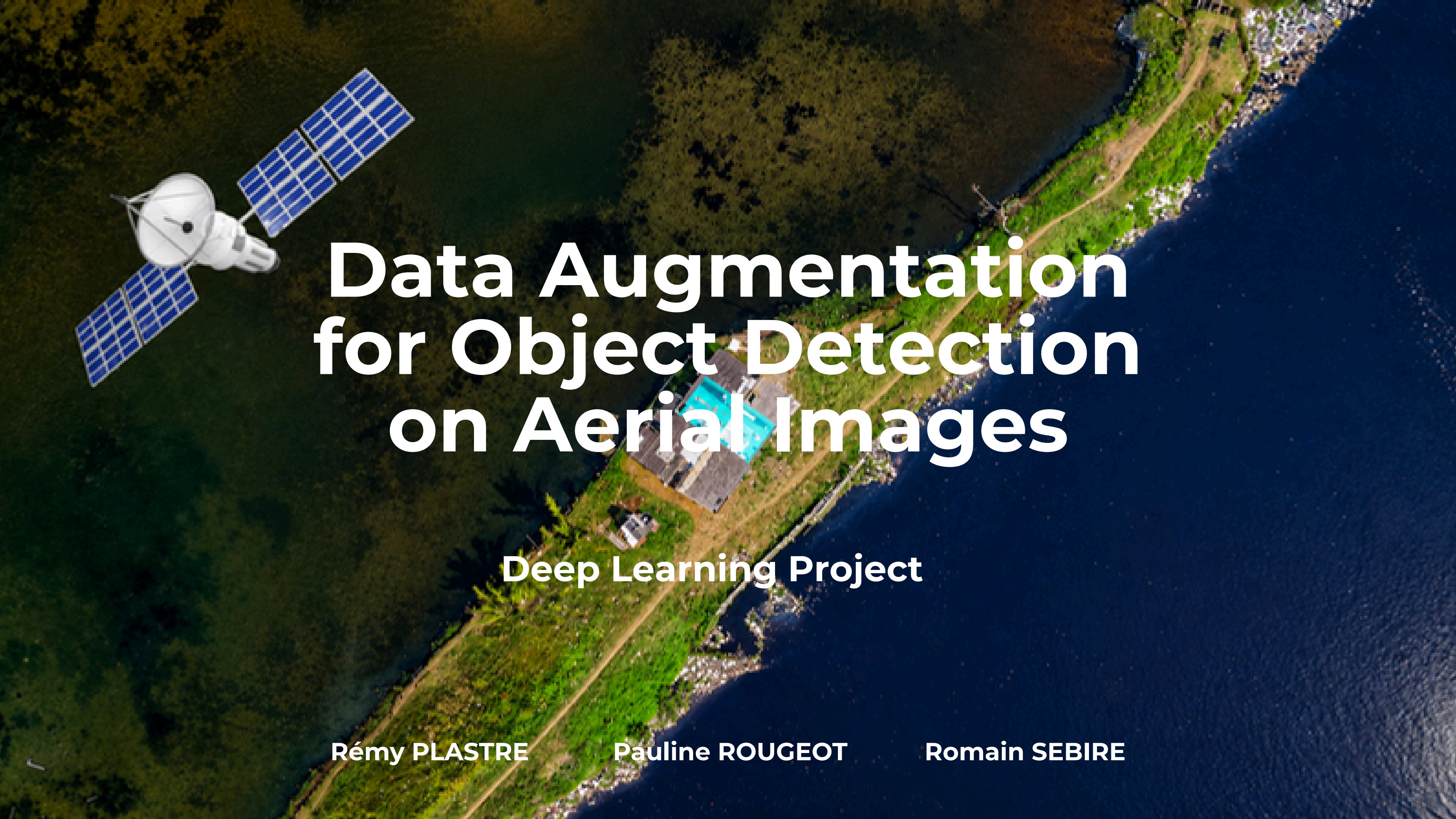


The background of the slide is an aerial photograph. In the upper left, a satellite with two large blue solar panel arrays is shown in orbit. Below it, a coastal landscape is visible, featuring a dirt road, green vegetation, and a small house with a bright blue roof. The right side of the image shows a dark blue body of water. The title text is overlaid on the central part of the image.

Data Augmentation for Object Detection on Aerial Images

Deep Learning Project

Rémy PLASTRE

Pauline ROUGEOT

Romain SEBIRE

The CADOT Project



CHALLENGE ON CITYSCAPE AERIAL IMAGE DATASET FOR OBJECT **DETECTION**

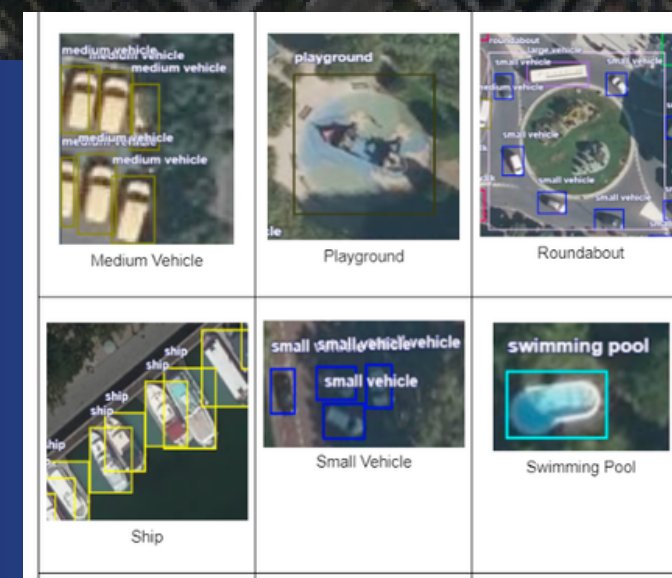
IEEE ICIP 2025 --- Grand Challenge

Congratulations to the winners!

First Place 🏆: Team Double J
Jing Jie TAN and Yi Jie WONG
Universiti Tunku Abdul Rahman, Malaysia

Second Place 🥈: Team HUS_ChapHet
Minh Hieu VU, The Son PHAN and Trong Duc NGUYEN
Hanoi University of Science, Vietnam National University, Vietnam

<https://cadot.onrender.com>



Cityscape Aerial Dataset for Object DeTection: **benchmark dataset**
4,628 images with over **100,000 annotations** across **14 urban object categories**

Our Goal

Explore all performances maximization

- “Challenge mode”
- **Benchmark** experimental techniques
- Maximize **rare-class detection**

In practice

1. Detect and classify **14 object categories**
2. **Finetune YOLO-based** architectures on **satellite** photography
3. Apply **augmentation techniques** to improve detection performance

Dataset Characteristics

- 4,628 high-resolution 500×500 pixels aerial images from the Paris region.
- 106,691 object annotations across 14 categories: vehicles, buildings, playgrounds...
- Complex urban scenes with occlusions and varying object sizes

Sources: National Institute of Geographic and Forest Information (IGN)



Format: "jpg"

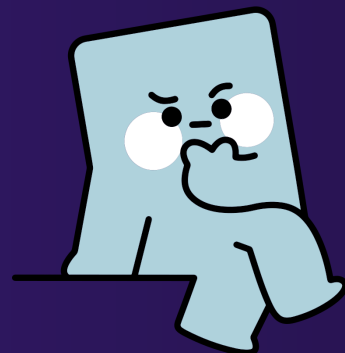


Annotations : COCO format annotation



Statistical Analysis

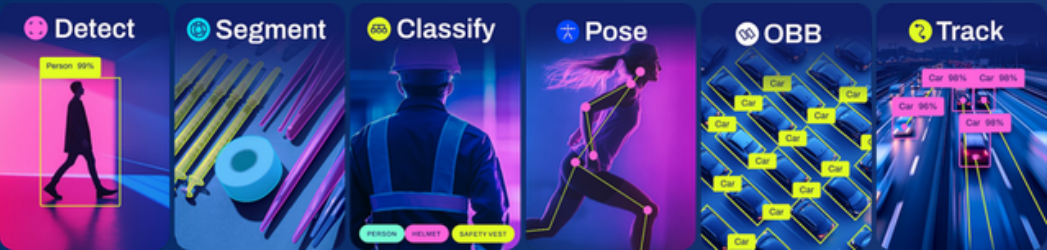
Class Name	basketball field	roundabout	playground	swimming pool	train	football field	tennis court
Count	15	37	44	60	67	77	98
% of Train Set	0.02%	0.05%	0.06%	0.08%	0.09%	0.10%	0.13%
Class Name	graveyard	ship	large vehicle	crosswalk	medium vehicle	building	small vehicle
Count	130	300	841	3751	3865	30764	35149
% of Train Set	0.17%	0.40%	1.12%	4.99%	5.14%	40.91%	46.74%



Major Issue : Extreme class Imbalance



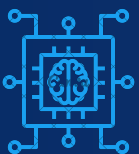
Yolo v11



Process images in a single pass
Optimal trade off between speed & accuracy



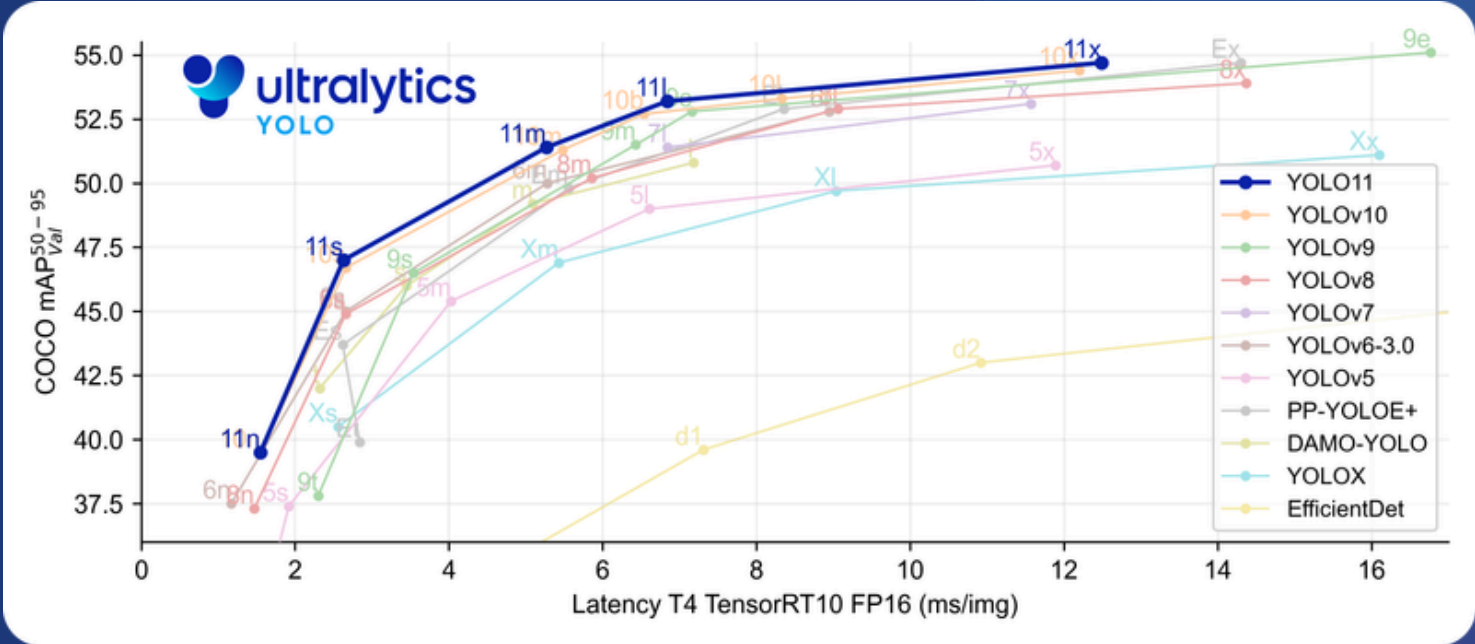
NANO



SMALL



MEDIUM



mAP^{val} 50-95	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	Parameters (M)	FLOPs (B)
39.5	56.1 ± 0.8	1.5 ± 0.0	2.6	6.5
47.0	90.0 ± 1.2	2.5 ± 0.0	9.4	21.5
51.5	183,2 ± 2,0	4,7 ± 0,1	20.1	68.0

(for object detection) <https://docs.ultralytics.com/fr/models/yolo11/#performance-metrics>



Evaluation metrics

Box(P)-precision

*It answers the question:
"When the model
predicts a Basketball
Field, is it actually one?"
High precision means
the model produces very
few "False Positives"*

R-Recall

*It answers the question:
"Did the model find all
the Basketball Fields
present in the image?"
High recall means the
model produces very
few "False Negatives"*

mAP50

*Prediction is considered
correct if the predicted
box overlaps at least
50% with the ground
truth box*

mAP50-95

*It rewards models that
not only find the object
but also fit the bounding
box perfectly around it*

Primary Metric
**"Precision & Recall in
single score"**

Baseline Results (mAP50)

Model/Class	basketball field	roundabout	playground	swimming pool	train	football field	tennis court
Nano	0.000	0.5227	0.000	0.520	0.122	0.152	0.327
Small	0.000	0.32	0.056	0.558	0.056	0.074	0.515
Medium	0.453	0.310	0.198	0.671	0.347	0.363	0.568
Model/Class	graveyard	ship	large vehicle	crosswalk	medium vehicle	building	small vehicle
Nano	0.393	0.578	0.410	0.793	0.569	0.684	0.793
Small	0.528	0.480	0.429	0.811	0.547	0.725	0.813
Medium	0.592	0.693	0.532	0.907	0.713	0.825	0.913



Model Choice

YOLO architecture was chosen for its ability to process images in a single pass (One-Stage Detector), offering an optimal trade-off between speed and accuracy.

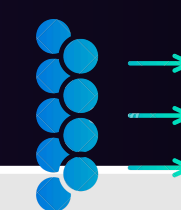
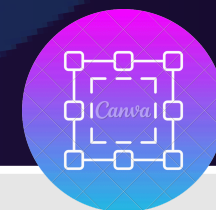
**FineTuning on CADOT
DataSet**

v11 Medium improved the global mAP to 0.578. However, **rare classes remained undetected**. The bottleneck was identified as data scarcity, not model architecture.

**How to deal with undetected
classes?**

Data augmentation !

D.A. : 3 methods compared



	Albumentations	Inpainting	Diffusion model
Definition	Classical data augmentation library applying geometric & photometric transforms with bbox support	Generates new objects by filling a mask on background images using an inpainting model + bbox reconstruction.	Fine-tunes SD1.5 with a LoRA on rare classes and generates synthetic samples for YOLO
Compute cost	Very light; GPU not required.	Moderate; diffusion inpainting is slower than classical aug.	High; requires GPU for LoRA training and sampling
Ease of Implementation	Easy; plug-and-play inside YOLO training pipeline.	Medium; requires mask generation, prompt tuning, and label updates	Hard; needs dataset prep, LoRA training, model integration, and re-labeling
Advantages	Fast, realistic augmentations; increases variability; stable bbox handling. No need for labelling	Highly realistic object insertion; preserves scene context; boosts rare classes.	Creates completely new, class-specific samples; excellent for rare objects; scalable.
Limitations	Cannot create new objects; limited to modifying existing pixels	Quality depends on mask detection & prompt; risk of artifacts; slower pipeline.	Training is costly; risk of domain shift; quality varies with LoRA training data

Method 1 : Albumentations



Key Features

- Fast: Optimized C++/NumPy backend for high-throughput training
- Robust: Large variety of geometric, photometric, and weather-based augmentations
- Detection-aware: Correctly transforms bounding boxes during augmentation
- Flexible: Easy to integrate into YOLO training

Types of Augmentations

- Geometric: flips, rotations, scaling, cropping, perspective
- Photometric: brightness/contrast, hue shifts, noise, blur
- Occlusion strategies: Cutout, CoarseDropout
- Distortions: elastic transforms, grid distortions

Albumentations

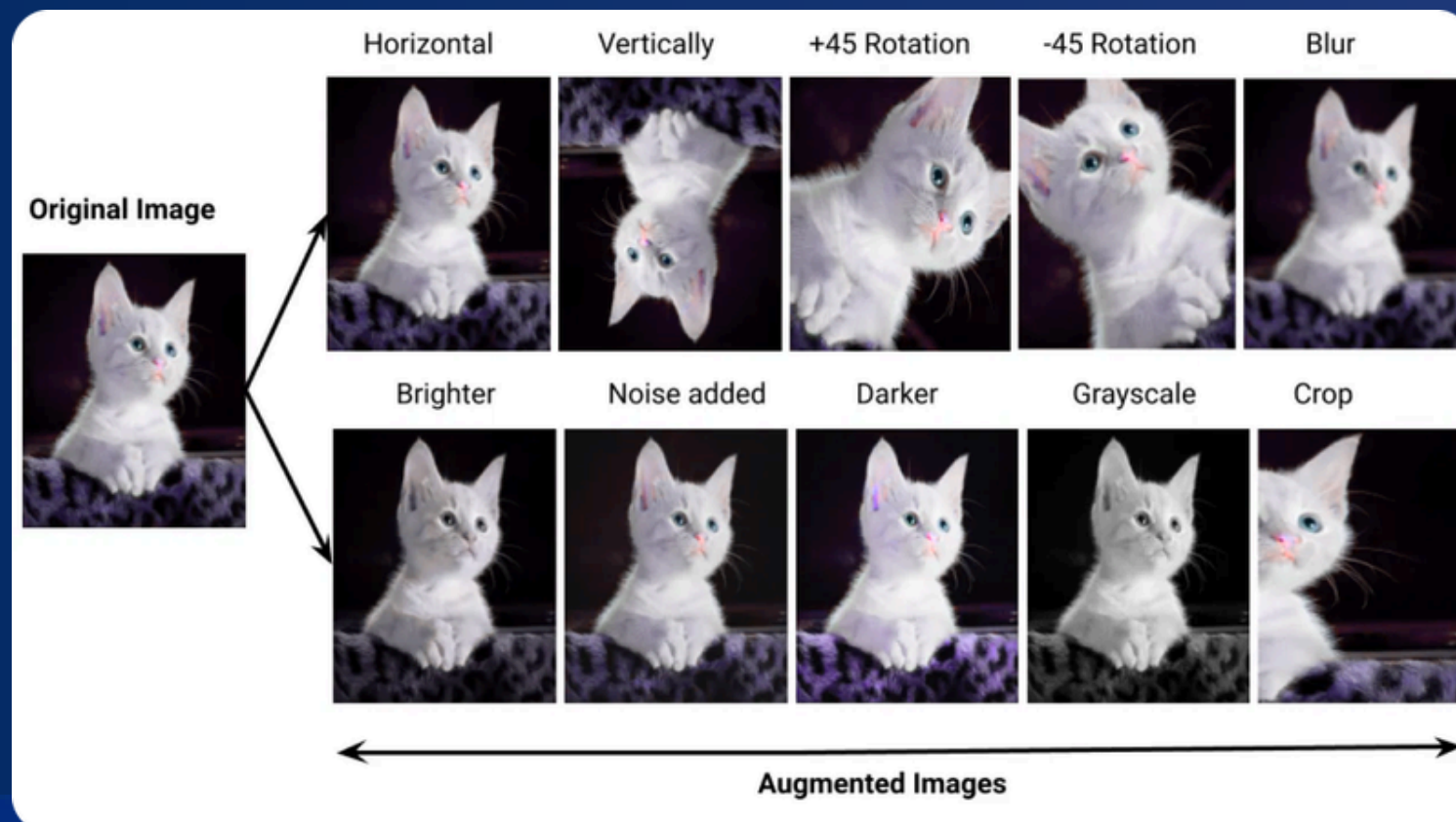
FIRST IDENTIFIED A LIST OF "RARE CLASSES" (IDS: 1, 4, 5, 8, 9, 13, 14).

Strategy

FOR EVERY IMAGE CONTAINING
AT LEAST ONE INSTANCE OF
THESE RARE CLASSES, IT
GENERATES **5 NEW** SYNTHETIC
VERSIONS

Pipeline

A SINGLE, UNIVERSAL PIPELINE
WAS APPLIED INDISCRIMINATELY
TO ALL SELECTED IMAGES



- **GEOMETRIC:** RANDOMROTATE90 (P=1.0), HORIZONTALFLIP (P=0.5), AND VERTICALFLIP (P=0.5) TO SIMULATE DIFFERENT DRONE/SATELLITE ORIENTATIONS.
- **PHOTOMETRIC:** RANDOMBRIGHTNESSCONTRAST (P=0.5) TO VARY LIGHTING CONDITIONS.
- **TEXTURE/QUALITY:** GAUSSIANBLUR (P=0.3) TO SIMULATE FOCUS ISSUES AND CLAHE (CONTRAST LIMITED ADAPTIVE HISTOGRAM EQUALIZATION, P=0.3) TO ENHANCE LOCAL CONTRAST.

Results

- **HIGH RECALL (0.638 VS 0.548):** THE MODEL SAW SIGNIFICANTLY MORE EXAMPLES OF RARE OBJECTS.
- **PRECISION DROP:** THE AGGRESSIVE USE OF A GENERIC PIPELINE (SPECIFICALLY GAUSSIAN BLUR AND HIGH CONTRAST CHANGES) ON ALL CLASSES DEGRADED THE VISUAL FEATURES OF SPECIFIC OBJECTS (E.G., BLURRING THE DISTINCT WHITE LINES OF SPORTS COURTS), LEADING TO INCREASED FALSE POSITIVES.

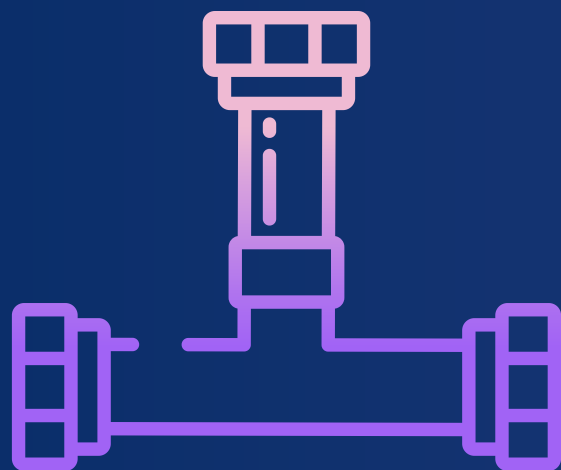
Albumentations

Strategy

INSTEAD OF APPLYING A
UNIFORM MULTIPLICATION
FACTOR, WE DEFINED
DYNAMIC COEFFICIENTS FOR
EACH RARE CLASS.

CRITICAL CLASSES (< 50 INSTANCES)

RARE CLASSES (50-150 INSTANCES)



A

THE "SPORT" PIPELINE (GEOMETRIC INTEGRITY)

- **TARGET CLASSES:** BASKETBALL, FOOTBALL, TENNIS, CROSSWALK.
- **OBJECTIVE:** THESE OBJECTS ARE DEFINED BY **RIGID BOUNDARIES** (WHITE LINES) AND SPECIFIC **GROUND COLORS** (CLAY, GRASS, ASPHALT). PRESERVING THESE HIGH-FREQUENCY DETAILS IS CRUCIAL.

B

THE "TEXTURE" PIPELINE (FEATURE ENHANCEMENT)

- **TARGET CLASSES:** SWIMMING POOL, GRAVEYARD.
- **OBJECTIVE:** THESE CLASSES ARE DEFINED BY INTERNAL PATTERNS RATHER THAN OUTLINES (E.G., THE RIPPLES OF WATER, THE REPETITIVE ARRANGEMENT OF TOMBSTONES). THE MODEL FAILED PREVIOUSLY BECAUSE IT CONFUSED THESE TEXTURES WITH GENERIC NOISE.

C

THE "SHAPE" PIPELINE (MORPHOLOGICAL VARIATION)

- **TARGET CLASSES:** ROUNDABOUT, PLAYGROUND, TRAIN, SHIP.
- **OBJECTIVE:** THESE OBJECTS HAVE DISTINCTIVE GLOBAL SHAPES BUT CAN VARY SIGNIFICANTLY IN APPEARANCE OR VIEWING ANGLE.

Results

- **GRAVEYARDS (+9.5%): THE TEXTURE PIPELINE SUCCESSFULLY TAUGHT THE MODEL TO RECOGNIZE TOMBSTONE PATTERNS.**
- **ROUNDAABOUTS (+8.8%): PRESERVING CIRCULAR GEOMETRY VIA AFFINE TRANSFORMS WAS CRUCIAL.**
- **LARGE VEHICLES (+9.3%): BETTER DISTINCTION FROM MEDIUM VEHICLES DUE TO REALISTIC VIEWING ANGLES.**

Method 2 : Inpainting

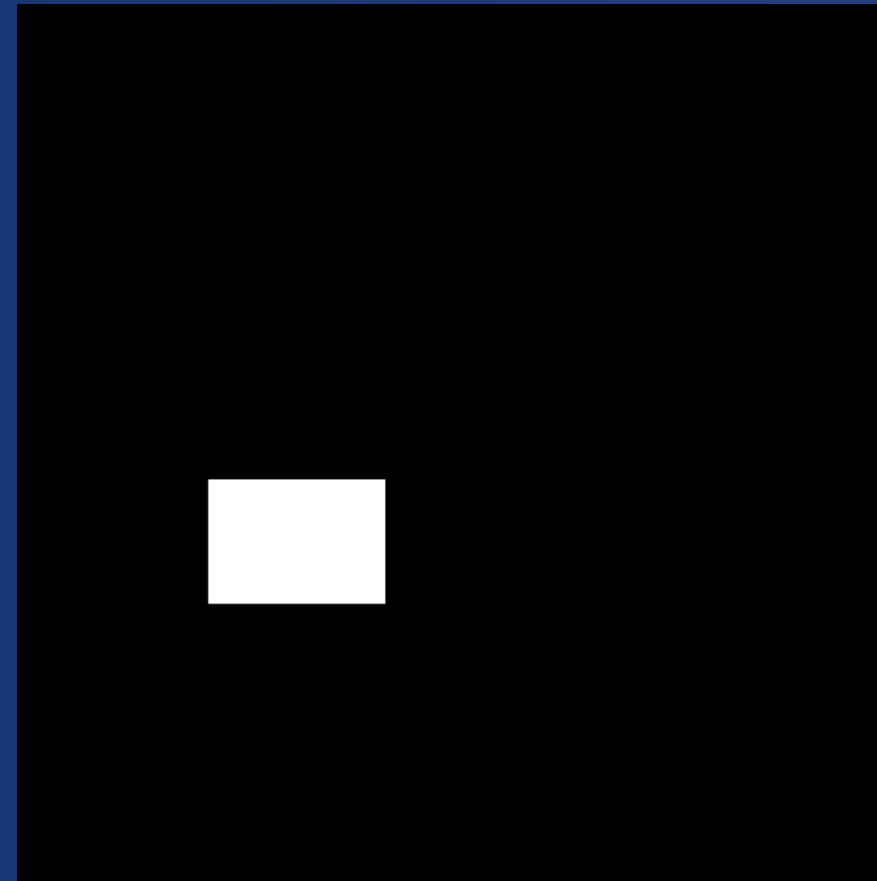


IMAGE SELECTION

MASKING

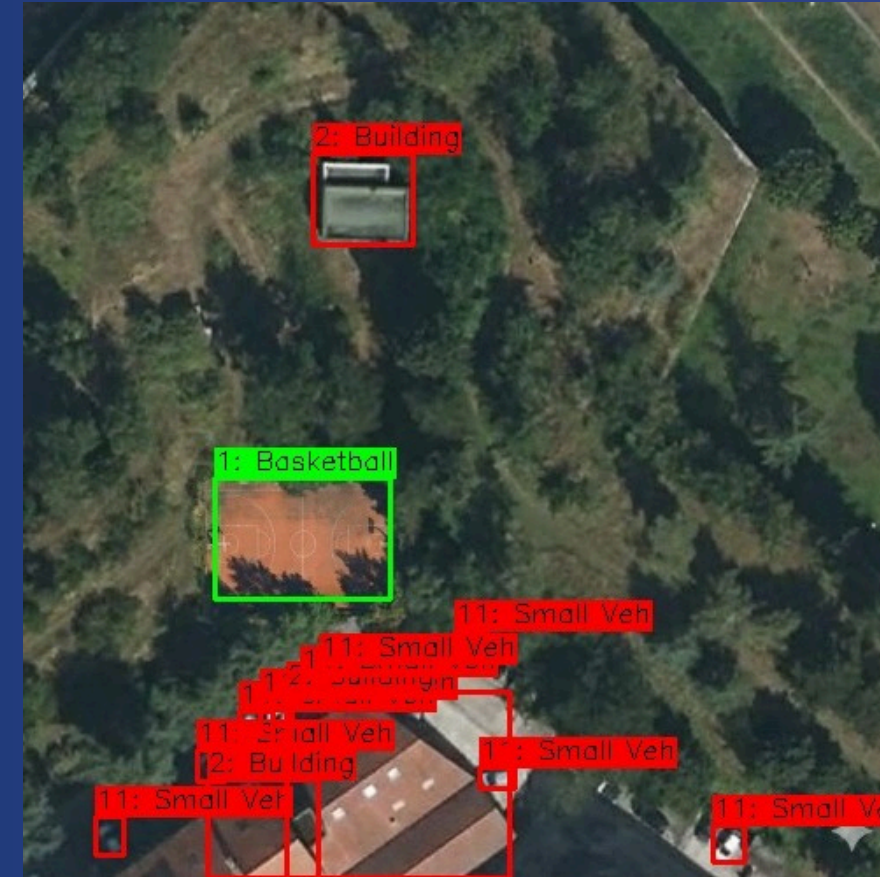
IMPAINTING GENERATION

EMPTY-REGION DETECTION

BOUNDING BOX STORED



Method 2 : Inpainting

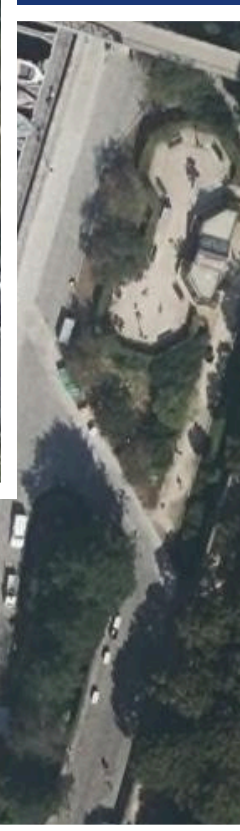
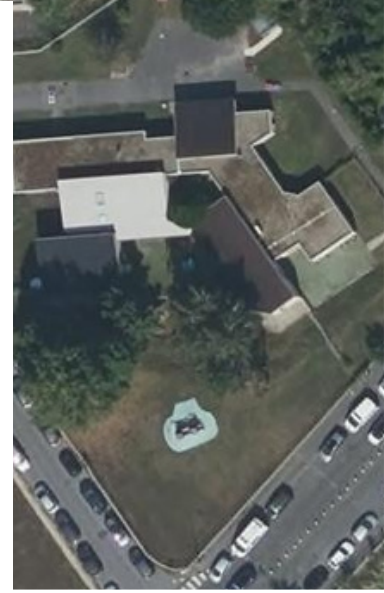
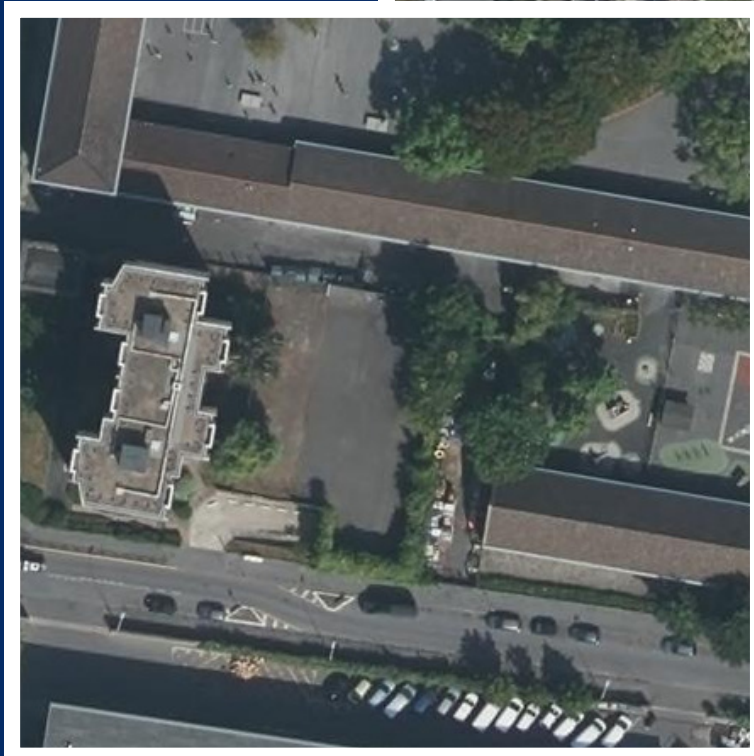
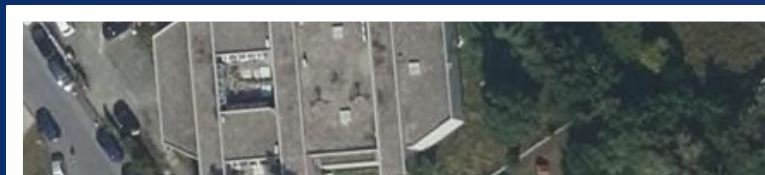
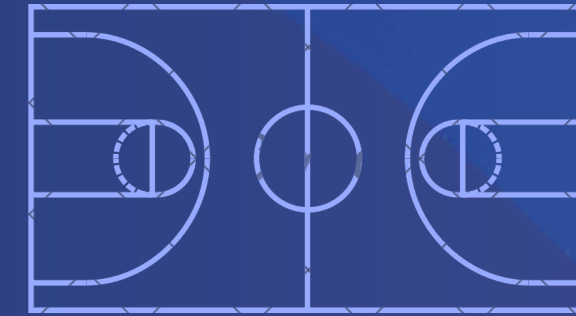


IMPAINTING GENERATION

NEW LABELED IMAGE

UPDATE LABEL

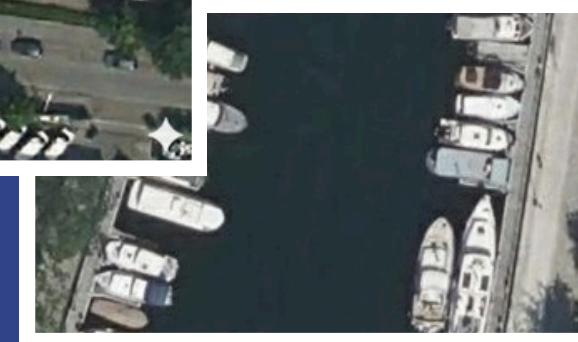
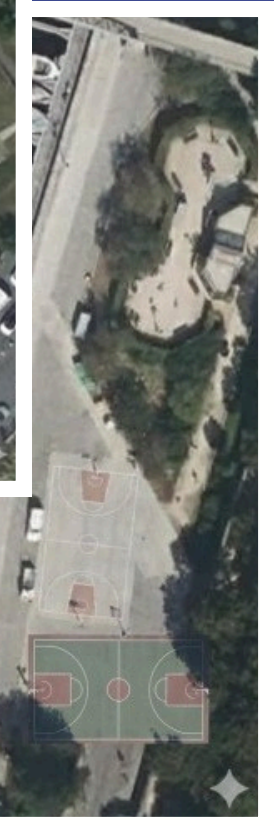
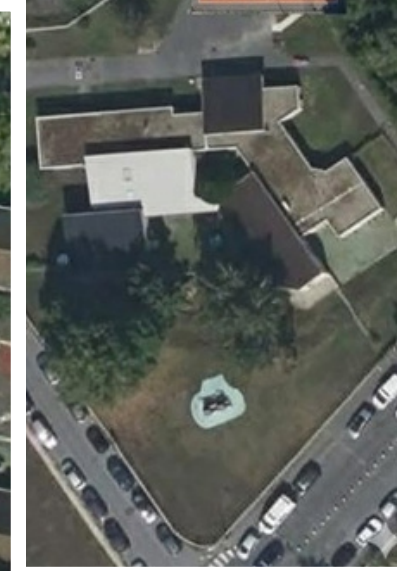
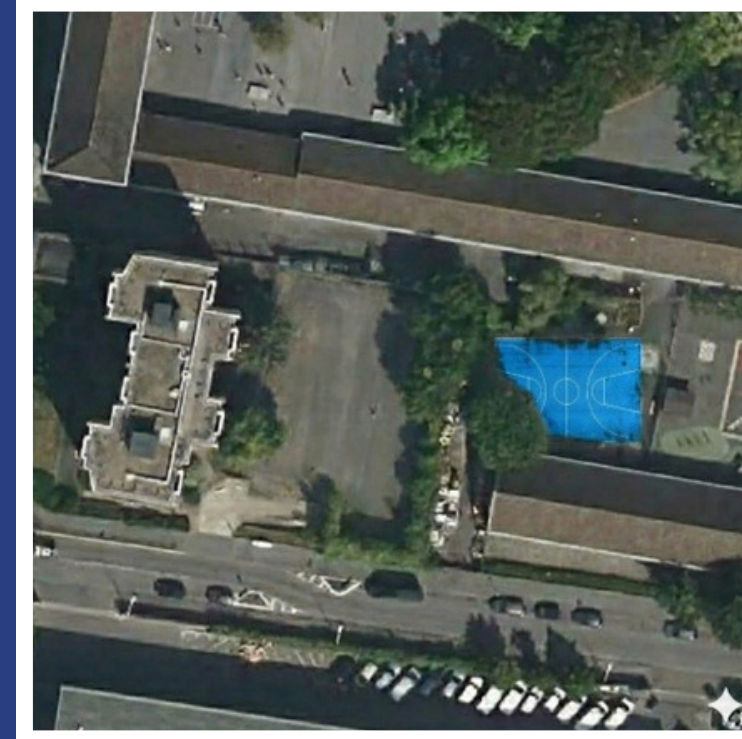
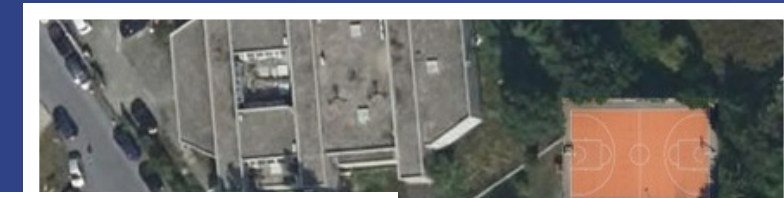
Method 2 : Inpainting



22 NEW IMAGES



BASKETBALL FIELD



Results

Model/Class	basketball field	roundabout	playground	swimming pool	train	football field	tennis court
Albu adaptative	0.195	0.398	0.081	0.611	0.485	0.439	0.525
Inpainting (Basket)	0.238	0.356	0.103	0.668	0.388	0.513	0.583
Model/Class	graveyard	ship	large vehicle	crosswalk	medium vehicle	building	small vehicle
Albu adaptative	0.687	0.748	0.625	0.897	0.726	0.824	0.907
Inpainting (Basket)	0.649	0.719	0.573	0.893	0.732	0.820	0.905

Recall for the Basketball class doubled (0.20 - 0.40)

Method 3 : Diffusion model

SD 1.5 Model + LoRA + ControlNET

- Training **UNet** on specialized basketball dataset (200), captions (textual descriptions) by LLM
- + **Cadot subset** filtered on basketball courts (13 original images → 200)
- Results : mAP 50 + 7.3% (0.264 → 0.337)



Roboflow dataset



CADOT dataset



SD Generation

Synthesis

What we achieved

- **YOLOv11 baseline**
- Three augmentation strategies: **Albumentations, Inpainting, Image generation**

Limitation

- Image generation without Coco annotation
- Difficulty to improve **all classes** at the same time
- **Low-rank adaptation** < complete SD **finetuning**

Future improvement

- **Hybrid strategy**: Inpainting & Generation
- **Specialised** model per classes

**Thank You For
Your Attention**